

To Whom it may concern,

With over 15 years of combined experience in academia and industry, in the field of computer science, I am passionate about contributing to an environment that fosters quality education and research, prepares students to navigate the challenges of a rapidly evolving technological landscape, and equips them for success in the job market.

Throughout my career, I have demonstrated a commitment to excellence in teaching and a passion for advancing the field of computer science. My portfolio encompasses many years of industry experience as a software programmer, where I consistently delivered products that exceeded client expectations. This foundation, combined with my academic journey—starting with a Teaching Diploma in Applied Mathematics followed by a Professional Master's and culminating in a PhD in Computer Science—has deeply influenced my teaching philosophy and methodology. I strive to produce students who are not only academically and theoretically strong but also well-prepared for the job market.

As a faculty member, I have taught a broad range of computer science courses, consistently earning positive feedback for my ability to demystify complex concepts and engage students in meaningful learning experiences. In addition to my teaching experience, my doctoral research focused on cybersecurity, static program analysis and programming language compilers, areas that have provided me with a deep and nuanced understanding of critical aspects of computer science. This research has not only enhanced my technical expertise but has also informed my pedagogical approach, allowing me to offer students a well-rounded and comprehensive education.

Beyond the classroom, I have extensive experience in the development of web and database applications, working with technologies such as Java SE, Jakarta EE, Spring Framework, .NET, C#, ASP.NET, MS SQL Server, and Crystal Reports. This practical experience enables me to bridge the gap between theoretical knowledge and real-world application, a skill I incorporate into my teaching to better prepare students for the demands of today's job market.

My academic background and research in cybersecurity, particularly in static program analysis and software development methodologies, have provided me with a solid foundation in both the theoretical and applied aspects of the field. My teaching experience at Maharishi International University has allowed me to refine my teaching strategies, focusing on student-centered learning that emphasizes active participation, collaboration, and real-world problem-solving. Additionally, I have been involved in curriculum development, contributing to the creation of new courses that address the latest trends in computer science and artificial intelligence.

Your commitment to excellence in education, diversity, and community service resonates deeply with my own values. I am dedicated to creating an inclusive and supportive learning environment where students from all backgrounds can thrive. I am excited about the possibility of contributing to this mission through my teaching, research, and community involvement.

I look forward to the opportunity to discuss how my background, skills, and experiences align with the needs of your institution and how I can contribute to your continued success. Please feel free to contact me if you require any additional information.

Sincerely,

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Assistant Professor of Computer Science

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Teaching Statement

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I hold a Teaching Diploma in Science from the Lebanese University, a Master of Science in Computer Science from Maharishi University of Management, and a Ph.D. in Computer Science from Oakland University. This diverse academic background has provided me with a strong foundation in both the theoretical and applied aspects of computer science, significantly shaping my path as an educator. Beyond academics, my industry experience has also enriched my teaching philosophy, emphasizing the importance of preparing students to meet real-world challenges. This preparation is essential and critically important in an applied field like computer science. It is this combination of my diverse academic background and extensive professional experience that has shaped my teaching philosophy and approach.

Teaching Philosophy and Approach

At the core of my teaching philosophy is the belief that education should serve as a bridge between theoretical knowledge and practical application. The role of the educator is to effectively communicate complex theoretical concepts to students with clarity. I am convinced that true mastery of a theoretical topic is demonstrated by the ability to explain it clearly to anyone, using the right words and examples. Driven by this conviction, I strive to make complex, abstract concepts accessible to students by relating them to real-world challenges and experiences. For instance, when teaching web programming, I draw on my expertise in research and cybersecurity to highlight the importance of secure coding practices. I build on my solid theoretical foundation in cybersecurity and my practical experience to provide my students with well-rounded knowledge, explaining security concepts and developing relevant examples that demonstrate the importance of security best practices in protecting distributed applications from malicious exploitation in today's digital landscape.

As an Assistant Professor of Computer Science at Maharishi International University, I have had the privilege of teaching a wide range of courses, including computer organization and architecture, low-level programming, fundamental and advanced Java programming, modern programming practices and web programming. I also teach cybersecurity, software engineering, DevOps, microservices, and CI/CD. My approach to teaching these subjects is student-centered, focusing on active learning, hands-on projects, and fostering a collaborative learning environment. I encourage my students to think critically, solve problems creatively, and apply their knowledge in practical settings.

My role as an educator extends beyond the classroom. I am deeply committed to mentoring and advising students, guiding them through their academic and professional journeys. I take pride in helping students identify their strengths, set goals, and navigate challenges, all while fostering a growth mindset. Whether through office hours, one-on-one meetings, or group discussions, I am dedicated to supporting my students' academic, professional, and personal development.

Curriculum Development and Continuous Improvement

The field of computer science and technology, whether at the level of hardware or software, is ever-changing. Staying at the forefront of technological advancements is essential to equipping my students with the latest and most relevant information for the job market. I am constantly researching software development methodologies, new technologies and current industry trends. Additionally, I actively participate in curriculum development to ensure that MIU's computer science program remains up-to-date with industry needs. My ongoing professional development includes performing research and attending conferences, engaging in online forums, and continuously learning about new technologies and methodologies. This commitment enables me to incorporate the latest tools and techniques into my courses, providing students with a cutting-edge education that prepares them for the rapidly evolving tech industry.

Conscious-Based Education and Student Engagement

Maharishi International University is the home of Consciousness-Based Education (CBE), with Transcendental Meditation (TM) at the core of this educational methodology. TM is a scientifically proven technique used as an active means of rest. It is well-known that a rested physiology promotes sharper thinking and results in a brain ready for action. MIU incorporates group meditation into the curriculum, where students and teachers meditate together in the classroom as part of regular classes. This methodology creates a positive and supportive learning environment, enhancing students' ability to focus, reducing stress, fostering a sense of belonging, and promoting self-motivation. It also encourages a desire in students to emulate their teachers and grasp knowledge more easily and effectively. CBE has significantly influenced my approach to student engagement. I integrate these principles into my teaching, helping students not only absorb technical knowledge but also develop essential soft skills such as teamwork, communication, and mindfulness. These skills are crucial for their success both academically and personally, contributing to their overall self-accomplishment.

Impact and Vision

My ultimate goal as an educator is to empower students to become knowledgeable, innovative, and ethical professionals who can contribute meaningfully to the field of computer science. I am passionate about nurturing the next generation of computer scientists who are not only technically proficient but also capable of addressing the ethical and societal implications of their work.

Looking forward, I am excited to continue refining my teaching methods, exploring new ways to integrate emerging technologies into the curriculum, and contributing to the academic community.

I am committed to making a lasting impact on my students, helping them achieve their full potential and preparing them to face the challenges and opportunities of the future.

Research Statement

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My research focuses on the intersection of cybersecurity and static analysis, a specialized area of computer science that is crucial for ensuring the reliability and security of software systems. I completed my Ph.D. at Oakland University in Michigan, USA. My dissertation concentrated on static program analysis of Java program code, with a particular focus on security vulnerabilities in mobile applications, especially those running on the Android platform.

Research Background and Contributions

Static program analysis is a powerful tool in cybersecurity. It enables the automatic detection of vulnerabilities within software code. Unlike dynamic analysis, static analysis does not require execution. This approach employs formal methods and mathematical models to identify potential flaws. It seeks correctness, which makes it invaluable in the ongoing effort to protect sensitive data, networks, and systems from cyber threats. My work has primarily focused on developing new, theoretically sound software analysis techniques, as well as enhancing the accuracy and performance of existing techniques. The ultimate goal is to improve the automated discovery of critical vulnerabilities. Automation, crucial for developing secure software, is essential in any software development methodology, especially in continuous integration and continuous delivery, where it ensures the security and reliability of software.

During my doctoral studies, I focused on static analysis techniques for Java programs, particularly applying these techniques to enhance the security of mobile applications. My work primarily focused on string analysis, numerical analysis, taint analysis, and command injection analysis. Taint analysis is used to trace the flow of sensitive information through a program, identifying paths where data might be exposed through public sinks. This method is crucial in preventing data breaches and ensuring that sensitive information is not inadvertently leaked. On the other hand, command injection analysis investigates how unsanitized input might be exploited by malicious actors, potentially leading to the execution of unauthorized commands with the same privileges as the host application. Value sensitivity was used to improve the accuracy of both analyses, reducing false positives and the need for manual intervention.

Research in static program analysis is highly theoretical, requiring a deep understanding of formal methods and their application to real-world problems. However, I have always prioritized the practical implications of my work, striving to bridge the gap between theory and application. This is demonstrated by accompanying my research papers with fully-fledged implementations of the analyses, made available as open-source projects. As mobile technologies have become integral to modern life, ensuring their security has become critically important.

Current and Future Research Directions

Since completing my PhD, my research has expanded to explore the integration of static program analysis within the broader context of software development methodologies, particularly DevOps and CI/CD. DevOps has become a critical methodology in modern software engineering, emphasizing automation to enhance the reliability and efficiency of software release cycles. My recent work has focused on investigating how static program analysis can be leveraged to ensure the security, verification, and assurance of code within DevOps pipelines. By automating the detection of vulnerabilities and verifying the correctness of code before deployment, static analysis can reduce the need for manual code reviews, thereby optimizing the software development process.

In addition to my work on static analysis in mobile applications, I have also investigated its application in distributed web programs, which share similar challenges related to lifecycle and control flow as those encountered in Android development. My goal is to develop tools that can automatically analyze and verify distributed systems, ensuring they meet security and performance requirements without the need for extensive manual intervention.

Looking ahead, I plan to further explore the role of static program analysis in emerging software engineering practices. This includes investigating how these techniques can be adapted to new paradigms such as microservices, serverless computing, and other distributed architectures that are becoming increasingly prevalent in industry. My goal is to develop innovative solutions that integrate static analysis seamlessly into the software development lifecycle, ensuring that security and reliability are maintained even as the complexity of systems continues to grow.

Impact and Vision

The impact of my research extends beyond the academic community, contributing directly to the development of safer and more secure software systems. By improving the tools and methods used to detect vulnerabilities, my work helps ensure that the software upon which society heavily relies is both reliable and secure.

Moreover, I am committed to continuing my contributions to scholarly communication through publications in high-quality, peer-reviewed conferences and journals. I believe that sharing knowledge and collaborating with others in the field is essential to advancing the state of the art in cybersecurity.

At the core of my research is a dedication to addressing real-world challenges. As I look to the future, I am excited to continue pushing the boundaries of what is possible in static program analysis, with the ultimate goal of creating a safer digital environment for all.

Diversity and Inclusion Statement

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As an educator with a diverse academic and cultural background, I am deeply committed to fostering an inclusive learning environment where all students feel valued, respected, and empowered to succeed. My experiences, ranging from earning a Teaching Diploma in Science from the Lebanese University to obtaining a Professional Master of Science in Computer Science from Maharishi University of Management, completing a Ph.D. in Computer Science at Oakland University, and gaining extensive industry experience, have profoundly shaped my understanding of the vital role diversity plays in education. I believe that the richness of diverse perspectives is essential to innovation and growth in the field of computer science.

My teaching philosophy is rooted in the belief that education should be accessible to all students, regardless of their background or circumstances. I strive to create a classroom environment that bridges the gap between theoretical knowledge and practical application, ensuring that every student can relate abstract concepts to real-world experiences that resonate with their unique perspectives. By using practical examples and culturally relevant content, I make complex ideas accessible to students from all walks of life.

In my role as an Assistant Professor at Maharishi International University, I have consistently worked to promote inclusivity in my teaching practices. I emphasize active learning, hands-on projects, and collaborative activities that encourage students to learn from each other's diverse experiences. My commitment to diversity extends beyond the classroom, as I am dedicated to mentoring and advising students, particularly those from underrepresented groups in computer science. I guide them through their academic and professional journeys, helping them navigate challenges, identify their strengths, and achieve their goals.

Maharishi International University's Consciousness-Based Education (CBE) has also played a significant role in shaping my approach to diversity and inclusion. The incorporation of Transcendental Meditation (TM) into the curriculum fosters a positive and supportive learning environment that enhances focus, reduces stress, and promotes a sense of belonging. These principles align closely with my commitment to creating an inclusive classroom where all students can thrive. By integrating CBE into my teaching, I

help students not only master technical skills but also develop the soft skills necessary for success in a diverse and global workforce.

I am also actively involved in curriculum development, ensuring that the courses I teach are inclusive and reflect the latest industry trends and technologies. I believe that staying current with technological advancements is crucial to providing all students with the tools they need to succeed in the rapidly evolving tech industry. By continuously updating the curriculum to include diverse perspectives and emerging technologies, I aim to equip my students with a cutting-edge education that prepares them for the challenges of the future.

In conclusion, my vision as an educator is to empower all students to become knowledgeable, innovative, and ethical professionals who contribute meaningfully to the field of computer science. I am passionate about nurturing the next generation of computer scientists who are not only technically proficient but also attuned to the ethical and societal implications of their work. I am committed to making a lasting impact on my students by fostering an inclusive environment that celebrates diversity and prepares them to succeed in a global and interconnected world.

Teaching Experience and Course Overview:

Course Title: CSI 5350 - Programming Languages and Compilers.

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2017, Fall 2018, Fall 2019, Fall 2020, Fall 2021.

Course Description: This interdisciplinary course bridges the fields of Compiler Techniques and Program Verification, focusing on advanced methods to ensure software correctness and safety. It equips students with the theoretical and practical tools needed to analyze, verify, and ensure the reliability of complex software systems, preparing them for both academic research and industry applications in software verification.

The course builds on traditional compiler techniques, such as data-flow analysis, control-flow analysis, type checking, and abstract interpretation, which form the foundation of modern program analysis. Students explore various approaches to program analysis and construct a compiler for a programming language, incorporating one of the techniques they have learned to analyze the program as part of their coursework.

Program analysis is increasingly relevant in industry because it enables the discovery of bugs and security vulnerabilities before the program is executed, through static analysis at compile time. Many platforms are adopting these techniques to automate the verification of third-party applications and ensure their safety on their platforms.

I have assisted in delivering this course numerous times, and the knowledge I gained through taking and assisting with the course has provided a foundation for my research. My responsibilities included grading assignments, running lab sessions, and helping students assimilate the material and complete their projects.

Role: Teaching Assistant / Instructor

Course Title: CSI 3430/APM 3430: Theory of Computation

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2019, Fall 2020.

Course Description: This course introduces students to formal models of computation, exploring key concepts in automata theory, formal languages, and computability. Students will learn how different computational models—ranging from finite state automata to Turing machines—can recognize various classes of languages and address fundamental questions about what can and cannot be computed.

Throughout the course, students will study Finite Automata and Regular Languages, gaining the ability to design and analyze deterministic and non-deterministic finite state automata (DFA/NFA). They will also learn to identify and describe regular languages and represent them using regular expressions.

Additionally, students will explore Pushdown Automata and Context-Free Languages (CFL). They will design pushdown automata (1-PDA) with one stack to recognize CFLs, deepening their understanding of this class of languages.

Students will also learn to explain the mechanics of a Turing machine and its variations. They will be able to differentiate between languages that are accepted, recognized, and non-recognized by Turing machines. Furthermore, the course will explore the limits of computation through discussions of computability.

This course equips students with the foundational concepts of computation theory, laying the groundwork for understanding the capabilities and limitations of computer systems.

Role: Teaching Assistant / Instructor

Course Title: CSI 5610 - Advanced Data Structures and Algorithms

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2019, Fall 2020.

Course Description: CSI 5610 provides an in-depth study of advanced data structures and the principles of algorithm design and analysis. The focus is on techniques for designing efficient algorithms using appropriate data structures, proving their correctness, and analyzing their computational complexity. Key topics include hash tables, data structures for combinatorial optimization, search trees, recurrence relations, divide-and-conquer strategies, dynamic programming, greedy algorithms, advanced graph algorithms, and linear programming. Additionally, the course incorporates algorithms from recent technical literature to explore current advancements in the field.

Role: Teaching Assistant / Instructor

Course Title: CS363 - Computer Organization.

Institution: Maharishi International University.

Term/Year: Winter 2021, Winter 2022.

Course Description: This 4-credit course introduces the internal structure of computers, covering topics such as assembly language programming, machine language, and the design of digital logic circuits. Students study Boolean algebra and logic gates, as well as essential

combinational and sequential circuits used in computer design. They learn the fundamentals of computer organization and architecture and gain hands-on experience programming in assembly language. Additionally, students explore key components of computer systems, including memory and the CPU, through topics such as flip-flops, decoders, multiplexers, registers, and sequential circuits.

Role: Lead Instructor

Course Title: CS-390: Fundamental Programming Practices

Institution: Maharishi International University

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2022, Spring 2023, and Fall 2023.

Course Description: The Fundamental Programming Practices (FPP) course provides students with a comprehensive understanding of the Java programming language, emphasizing key object-oriented programming principles, data structures, and algorithm development. This course is designed to address foundational gaps students may have as they begin their MSCS program, ensuring they acquire the essential skills needed for advanced study.

Role: Lead Instructor.

Course Title: CS 401 - Modern Programming Practices

Institution: Maharishi International University

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2021, Winter 2022, Spring 2022, Fall 2023.

Course Description: The **Modern Programming Practices (MPP)** course builds on the foundational concepts covered in the FPP course, advancing students' proficiency in the Java programming language. This course equips students with advanced programming skills, preparing them to design and develop complex software solutions using modern Java practices. In this course, students explore advanced OOP concepts such as inheritance, interfaces, polymorphism, and the Open-Closed Principle to enhance modular and maintainable code. They are also exposed to more sophisticated programming techniques, including collection processing, lambdas and streams, annotations, Java generics, and multithreading.

Role: Lead Instructor

Course Title: CS489 - Applied Software Development

Institution: Maharishi International University.

Term/Year: Spring 2023

Course Description: CS489: Applied Software Engineering offers students a comprehensive exploration of the software development lifecycle, from conception to deployment, with a focus on enterprise-level solutions. This course emphasizes core principles, best practices, and modern tools and technologies for building robust, high-quality software, particularly using the Java platform. In this course Students learn how to analyze requirements, design appropriate architectures, implement solutions, test code, and deploy software to production environments. The course emphasizes real-world skills such as web development, microservices, and cloud-native application development, along with software security.

Role: Co-Instructor

Course Title: CS57x Practicums in Computer Science

Institution: Maharishi International University.

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2021, Winter 2022, Spring 2022, Fall 2023, Winter 2024.

Course Description: CS57x offers students a valuable opportunity to apply and expand their academic knowledge in a professional setting. As part of the MUM ComPro Master's program, the practicum involves practical work experience through CPT (Curricular Practical Training), allowing students to bridge theory and real-world applications. By the end of the course, students must submit a 1,000-word STAR report, summarizing their experience and achievements using the STAR (Situation, Task, Action, Result) framework. This report prepares students to articulate their professional experiences in future job interviews.

The instructor for the CS57x Practicum course is the student's faculty advisor from the Computer Science Department. Their role is to guide students through the practicum experience, ensuring that students apply their academic knowledge effectively in a professional setting. The instructor provides feedback and support throughout the course, helping students reflect on their practical experiences. Additionally, the instructor evaluates the final report, ensuring that students articulate their learning outcomes in alignment with the course's goals.

The course is graded on a Pass/No Pass (P/NP) basis, with the STAR report serving as the primary assessment. The STAR report, structured around Situation, Task, Action, and Result, is evaluated using a detailed rubric. The rubric assesses the completeness and clarity of each section, as well as how effectively students connect their practical experiences to the concepts and skills gained from their coursework. Each section is weighed differently, with particular emphasis on the actions taken and their outcomes. The instructor totals the points from the rubric to determine whether the student meets the criteria for a passing grade.

Role: Co-Instructor